

Leheng Zhao (赵乐恒)

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EDUCATION

Fudan University, Shanghai, China

2024 - Present

B.Eng, College of Computer Science and Artificial Intelligence

Weiyu High School, Shanghai, China

2021 - 2024

High School Diploma

RESEARCH INTERESTS

My primary research goal is to build **generalizable embodied intelligence systems** that can perceive, reason, act, and adapt in the physical world. I am particularly interested in **robot learning and control**, **world models for robotics**, **reinforcement learning** and **data-efficient learning methods for physical reasoning**. In the long term, I hope to contribute to the creation of reliable, socially intelligent, and general-purpose robotic agents — systems that move beyond task-specific automation toward the kind of embodied intelligence envisioned in **Asimov's R. Daneel Olivaw**.

RESEARCH EXPERIENCE

Pair-Biased Token Transformer for Depth-Based

2025 - Present

Robot World Modeling

Mentor: Dr. Zhiao Huang, Prof. Hao Su

- Designed a token-based world-model architecture that unifies depth patch tokens, robot link tokens, and joint-state qpos tokens within a pair-biased transformer for structured robot-scene representation learning.
- Introduced pair-wise relational biases to encode both robot kinematic links and spatial relations among scene tokens, enabling the model to reason over robot geometry, object layout, and physical interactions in a shared latent space.
- Developed multi-head prediction objectives for next depth patches, next robot link/qpos states, and reward estimation, allowing the learned dynamics model to support both visual and proprioceptive state prediction.
- Integrated the learned world model with a pure CEM-MPC planner, where candidate action sequences are rolled out through the model and selected according to predicted long-horizon rewards without using a separate policy head.

HONOURS AND AWARDS

- Outstanding Thesis Award, Ministry of Education Talent Program**
- Silver Medal of the 34th Shanghai Youth Science and Technology Innovation Competition**